Task-Conditioned Residual Skills for Reliable Multi-Stage Contact Manipulation

An Evidence-Driven Research Manuscript

ARES Research Working Group

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Abstract

Multi-stage contact manipulation requires controlling object motion, orientation and tool contact beyond intermediate position attainment. We investigate constrained parameterized residual reinforcement learning in a 16-environment JAKA testbed using Isaac Sim 5.1 and Isaac Lab 2.3.0. Historical ordinary residual PPO improves aggregate unseen-geometry completion over feedback, but most improvement comes from the foundational pushing task rather than guided-slot insertion (P2). After preserving that evidence and a negative effect-history pilot, three bounded physical research iterations test contact-target authority, shared rotated-contact geometry with tool yaw, and a stateful re-contact executor. The last iteration compares ordinary and execution-active Gaussian PPO under the same 87-dimensional observation, five-dimensional action interface, controller, reward and interaction budget. A saved-mask objective demonstrably changes actor score gradients on identical real rollout samples. However, three independently trained policies obtain 22/25/20 versus 24/24/18 accepted P2 completions out of the same 32 development cases; shared strong feedback obtains 26/32. The paired gains of +2/-1/-2 do not support a consistent method advantage. The implementation isolates a real actor-update mechanism, not a new marginalization theorem or validated recovery policy. Actual paired videos, process traces and independently audited outcomes expose both physical successes and failures. Development scenes are not a sealed final test, and state-based synthetic manipulation is not field or real-robot validation. A publishable novel-method claim remains unsupported.

Keywords: contact manipulation; residual reinforcement learning; stateful skill parameters; execution-aware policy objectives; reproducible robotics.

Evidence status. True online PhysX training and actor-gradient isolation are verified. The new three-seed P2 objective does not consistently improve task completion or outperform shared strong feedback. No sealed-final-test, visual-policy or hardware-safety claim is made.

1 Introduction

Industrial robot skills are often implemented as a combination of geometric planning, feedback control and task-specific execution contracts. These components can provide reliable approach and transport motions, while sustained object contact remains sensitive to object motion, orientation, friction and tool placement. A locally successful preparation motion need not leave a state from which the next operation finishes stably. Reinforcement learning (RL) offers a way to adjust executable actions using physical task feedback without requiring it to replace inverse kinematics or collision-aware planning.

Our application context is an existing industrial skill architecture, ARES/ARES-R. The present algorithmic experiments, however, are deliberately narrower than that system: a single JAKA arm manipulates a free synthetic object with a research paddle in two consecutive contact stages. They are not claims of tray grasping, production cabinet operation or end-to-end autonomous field execution. Controlled October field records are retained on a separate reference track and are excluded from this public package.

Previous experiments already show that residual PPO can learn these tasks. They also expose an important methodological risk. Aggregate performance on a mixture of tasks can hide a weak result on the more contact-sensitive task, and an auxiliary network or learned switch can update without improving physical outcomes. On unseen geometry, residual PPO improves the aggregate result over a tuned controller, but most of the gain is attributable to P1. Simple waiting at handoff and learned holding do not deliver an independent advantage. These negative findings motivate a bounded investigation rather than further scaling of a weak mechanism.

We focus on the following question: *Can physically actionable contact parameters and a policy objective aligned with their actual execution improve multi-stage manipulation beyond matched ordinary PPO and strong feedback control?* This does not require showing that RL is indispensable. It requires measurable, repeatable gains under equivalent physical permissions, observations and interaction budgets. Execution history is one tested information pathway, not a mechanism that must succeed by assumption.

The supported deliverables are an auditable contact-RL experimental basis, the historical residual-control comparison, three actual closed-loop research iterations, and same-sample actor-gradient isolation for a stateful execution objective. They do not establish a superior new algorithm: the final three-seed primary-success differences change sign, and feedback remains stronger. A single-seed fixed-refresh time signal is retained without

promoting it to a general claim. Merely concatenating context, rotating coordinates, training a predictor or applying standard marginalization is not considered sufficient novelty. Sealed evaluation and broader method validation remain future requirements, not completed results.

2 Related Work and Novelty Boundary

Parameterized actions and skill composition. RAPS studies parameterized action primitives as an RL action interface [3]. STAP uses task-agnostic skill values and planning to reason about geometric dependencies in skill sequences [1]. These precedents rule out claims that parameterized skills or successor feasibility are themselves new. The question here is narrower: whether observed sustained-contact effects can improve a constrained residual decision in a physically continuous task, beyond ordinary residual RL and task-relative control.

Residual and geometry-aware skills. Residual Skill Policies already learn residual corrections in skill space [7]. HACMan++ studies contact-rich manipulation through actionable contact/action abstractions [6]. GRN addresses feasibility-related reasoning [2], while Fabrica combines planning and geometrically structured residual skills [10]. Therefore, using a nominal controller, task frame, geometry input or feasibility mask cannot independently support novelty. A successful extension would need to show that a particular execution-feedback pathway changes actions and improves P2 under matched resources, with competing explanations removed.

Recovery and heterogeneous composition. RecoveryChaining combines nominal policies with learned recovery [11]. TASP studies the composition of heterogeneous black-box skills [4]. Our experiments do not yet implement a high-level recovery scheduler or reproduce these benchmarks. A future recovery mechanism must start from genuine recoverable physical states and be compared with legitimate retry and re-contact controllers.

Implications for this manuscript. The supplied literature comparison documents mechanism overlap, available implementation evidence and replication costs. The empirical boundary is strict: existing ARES results are self-built simulation experiments, not official benchmark scores. The task-frame change is an engineering/inductive-bias baseline. Only a reproducible increment over that baseline, linked to a genuine outcome-conditioning intervention, could justify a stronger method claim. None of the present text claims the first use of residual RL, skill chaining, successor value or learned termination.

Source-level mechanisms and temporal execution. A bounded follow-up review pins HACMan++, Fabrica and RecoveryChaining to exact commits, reads their action execution and selected training modules, and records source hashes and licenses separately. HACMan++ separates contact grounding, approach and post-contact motion, but its virtual joint writes and robot resets are not imported into our continuous physical environment. Fabrica implements path-relative residuals and a held-part attachment assumption; these are not evidence of free-object contact grasping. RecoveryChaining distinguishes actual nominal-suffix execution from its predictive Lazy alternative; a predicted value is not counted here as a physical success. FiGAR already jointly selects actions and repetition durations [9]. Thus parameter holding or contact-event refresh is not a first temporal abstraction. Our candidate requires a matched stateful-executor comparison and independent task benefits, not novelty by renaming persistence or unused-parameter marginalization.

3 Problem Formulation

We consider a finite-horizon controlled physical process with robot state $q, \dot{q}$, free-object pose x_o, R_o, object velocities v_o, ω_o, current stage k, task geometry g, and a skill contract C. The contract describes permissible action dimensions, reference-motion bounds, workspace, orientation requirements and task termination. The simulator advances at 120 Hz and control actions at 60 Hz. Stage transitions preserve robot, object and contact dynamics; they do not reset the object to an advantageous state.

Observation and information. The historical actor receives a 56-dimensional state-based observation including robot state, object pose and velocities, object–tool relation, stage and subsequent target errors, a simulated net-contact proxy and known geometry/dynamics context. Exact friction and mass context are assumed available in simulation. This is not a learned visual observation or an unknown-parameter identification result. Future suffix success and independent acceptance labels are never actor inputs. A proposed feedback history h_t may contain only already observed execution effects.

Task-relative action. Let T_g express the local task frame and ξ_{nom} a nominal contact reference increment. A residual parameter policy yields

$$\xi_t = \mathcal{P}_{C_t}(\xi_{\text{nom}}(o_t, C_t) + f_\theta(o_t, C_t, C_{t+1}, h_t)), \quad (1)$$

where $\mathcal{P}_{C_t}$ applies executable bounds to the *combined* command. A DLS controller maps the reference to constrained joint commands. Joint soft limits, tool orientation and reference lead bounds remain active for both nominal and learning components. Task-frame conversion alone is baseline B2, not the proposed innovation.

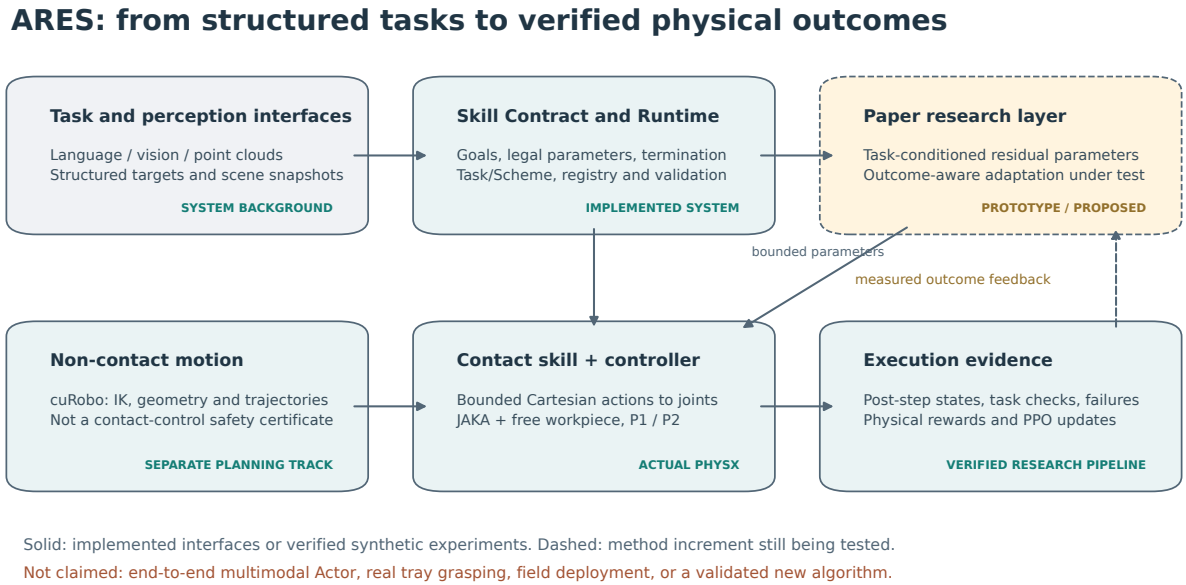


Figure 1: Scope of the study within the ARES skill architecture. The contact-RL experiment is implemented; multimodal policy deployment and long-horizon interfaces are not established results.

Objective and endpoint. The primary endpoint is independently accepted complete P2 success, not reward or intermediate position alone. The v6 contract requires target error below 8 mm, speed below 10 mm/s, angular speed below 0.2 rad/s, square-object symmetry yaw error below 8°, upright error below 5°, object height within 8 mm of the supported reference height, and 0.5 s of continuous stable completion, within a 12 s deadline. P1 is reported separately as a foundational task. A sampled geometric/process audit complements the environment's success flag.

The historical reward was fixed before its formal comparison:

$$r_t = 60\Delta d_t - \Delta t - 0.0002\|a_t\|^2 + 3I_{\text{switch}} + 10I_{\text{success}} - 5I_{\text{violation}}. \quad (2)$$

Progress is recomputed within each stage without counting a reset target distance as instantaneous achieved progress. This stagewise shaping is *not* claimed to be a globally policy-invariant potential difference: a phase-dependent potential would require an explicit jump correction. The new pilot uses the same stagewise progress and terminal/time terms across all learned controls, with action cost based on the applied reference rather than an unexecuted proposal. Reward changes are not an algorithmic contribution. Potential-based shaping $\gamma\Phi(s_{t+1}) - \Phi(s_t)$ remains an untested alternative. Current contact policies use fixed control intervals and ordinary MDP PPO; unequal-duration high-level options would require explicit SMDP discounting and are not asserted here.

4 Implemented Skill Interfaces and Candidate Objectives

4.1 Historical nominal control and residual PPO

The damped nominal controller tracks a stage target while respecting a fixed legitimate pushing side. Reference lead over the actual tool is bounded to 6 mm to prevent integral windup. Ordinary residual PPO adds at most 0.6 mm per control step in planar reference corrections. All methods share the same DLS joint controller and physical constraints. This strong nominal policy is retained and improved alongside any task-frame transformation; it is not weakened to manufacture an RL advantage.

The historical learning comparison uses 16 environments, a [64, 64] MLP, rollout length 256, six minibatch epochs and $\gamma = 1$. Each main model receives 65,536 online interactions. Final checkpoints are fixed by interaction budget, not selected on the frozen tests. PPO optimizes the usual clipped surrogate with GAE. A separate diagnostic mixes GAE with a current-rollout, value-bootstrapped $\lambda = 1$ target; it does not reuse stale episodes across policy updates and is not presented as a new credit-assignment method.

For clarity, the policy update is ordinary PPO [8], with likelihood ratio $r_t(\theta) = \pi_\theta(a_t|o_t)/\pi_{\text{old}}(a_t|o_t)$:

$$L_{\text{actor}} = \mathbb{E}_t\left[\min\{r_t\hat{A}_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\hat{A}_t\}\right]. \quad (3)$$

Context or history changes this distribution's inputs, not the estimator into a new credit-assignment algorithm. The value loss and optimizer use the same data budget as their matched baselines.

4.2 Historical v7 task-frame baseline B2

The new development comparison converts task errors, velocities, object orientation and permissible reference directions to the local entrance frame. B1 retrains the BODY-XY residual on the same new physical cases; B0 is the repaired task-local controller. B1/B2 share source physical state, input size, capacity, physical action range, reward and interaction budget, but exact information equivalence is not guaranteed. In particular, B2 canonicalizes the explicit entrance-direction cosine/sine to 1/0 while joint state and physical workspace remain in BODY coordinates. This is neither a strictly invertible whole-state transform nor a demonstrated rigidly equivariant policy. Any B1/B2 difference is a bundled engineering/representation comparison, not an isolated proof of frame causality. B2 and B3 retain the same base representation, supporting the targeted history ablation.

4.3 Historical v7 outcome-history candidate B3

The implemented B3 pilot adds eight one-step history channels to the historical 56-state basis: task-forward/lateral object finite-difference velocity (two), previous applied reference increment divided by control duration (two), their differences (two), object angular velocity about the vertical axis (one), and the previous normalized phase-0 contact-target command (one). The last quantity is a requested parameter, not an observed workpiece displacement. Each channel is bounded in observation preprocessing. These signals directly enter the actor; no future success label, learned predictor or disconnected auxiliary loss is used.

All learned pilot methods have three continuous outputs: planar residual correction bounded to 0.6 mm/control step and a phase-0 lateral contact-target offset bounded to 4 mm. Only the tool reference changes, not the physical object or task goal. B1/B2 set the extra eight history channels to zero but retain the same 64-dimensional input and network. They still observe current pose, velocity and contact proxy, so this ablation is called *no effect-history*, not absence of all physical outcome information. The downstream-goal ablation masks phase-0 future-target and some entrance-geometry channels while retaining mass/friction context; the legal action mapping still uses the known task direction. It is a *partial* next-contract intervention, not a policy with no downstream information.

Writing R_g for the planar task rotation, the measurable response part is formed as $R_g^\top (x_{o,t} - x_{o,t-1})/\Delta t$ and $R_g^\top \Delta x_{\mathrm{ref},t-1}/\Delta t$, together with their difference. The requested contact parameter is $4\,\mathrm{mm}\ \mathrm{clip}(a_{t,3}, -1, 1)$ in phase 0. It changes the controller's lateral contact target while the physical workpiece goal remains unchanged; rear-face alignment and reference antiwindup can suppress its instantaneous physical effect. A nonzero parameter command alone is not executed displacement. A cold-start paired probe records actual applied references and object trajectories to test whether it affects subsequent results.

This is ordinary full-chain-reward PPO with a real parameter-action interface. It does not implement a learned successor Q-function or learned stage termination, and history concatenation alone is not a new algorithm. Its scientific status is a prototype whose independent benefit must be tested. The actual run-source snapshots and protocol define the implementation; a single identified pre-fix observation-index error is preserved as diagnostic data and excluded from the corrected formal pilot.

4.4 Falsifiable ablations

The mechanism comparison removes outcome history, removes the next contract, provides capacity-matched ineffective conditioning, or trains a predictor without connecting it to policy decisions. These controls are required to separate information, representation, extra data and actual mechanism effects. Not-yet-run cells remain explicitly untested. A finite search or supervised scorer is included only when the implemented parameter decision has a well-defined candidate interface; a one-shot choice is labelled a contextual bandit rather than long-horizon RL.

4.5 Subsequent execution-aware parameter interface

The R3 protocol replaces ineffective history concatenation as the primary candidate with a shared stateful contact executor, not a new success criterion. Its five proposals are planar residual XY, phase1 lateral contact offset (8 mm bound), tool yaw (6 degree command bound), and phase0 contact-normal offset (40 degree bound). The formal input has 87 dimensions, including held parameters, timer, re-contact mode, pre-action mask, reset object XY and peak displacement needed by the recovery trigger; an earlier 84-dimensional smoke is kept separate. A continuous controller executes bounded withdrawal, planar bypass and re-approach without resetting the workpiece. This helper is not a complete robot or rail collision planner. All comparison policies receive the same executor and 24 s task limit; extra time or geometric repairs cannot be assigned to the learning objective.

Parameters persist for at most 32 control steps. An already observed force-proxy Boolean transition, angular-velocity change or lateral-velocity change can request earlier refresh after an eight-step minimum. This is contact-event heuristics, not a validated full contact classifier. During ordinary control, XY proposals execute; phase-specific parameters update only when legal and refreshing. During automatic re-contact even XY proposals are inactive. Held values are state memory, not fresh Gaussian samples.

Given a mask m_t computed before sampling and saved with the rollout, the candidate evaluates

$$\ell_{\mathrm{exec}}(a_t, s_t) = \sum_{i=1}^{5} m_{t,i} \log \mathcal{N}(a_{t,i}; \mu_{\theta,i}, \sigma^2_{\theta,i}). \qquad (4)$$

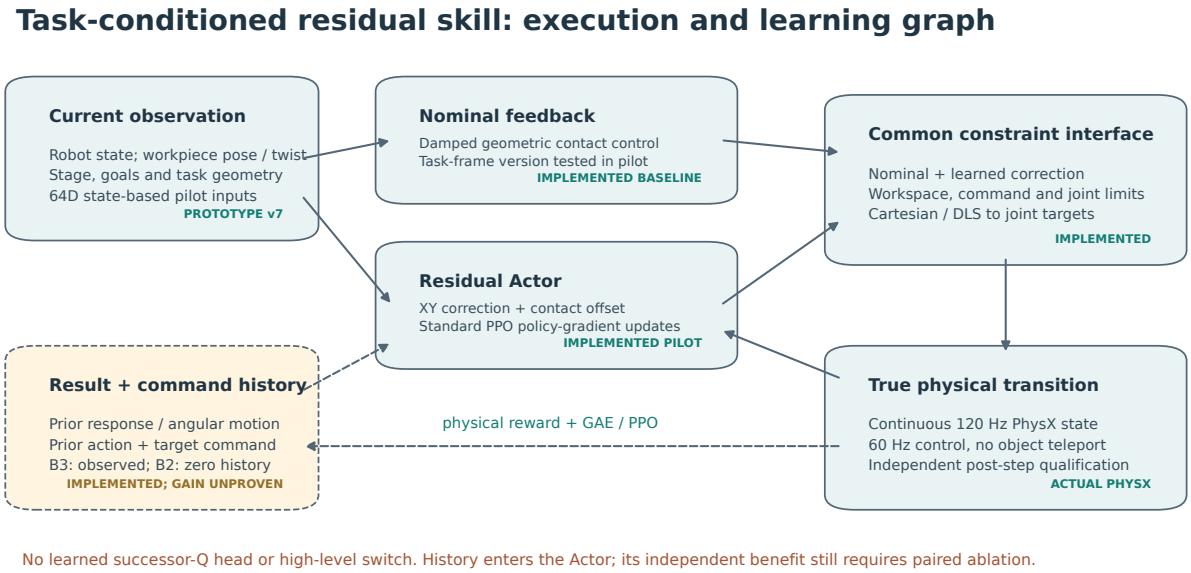


Figure 2: Historical v7 computation and evidence flow (64 observation/three action dimensions), not the later R2/R3 executor. The outcome-history prototype enters the actor, but a performance contribution requires matched physical ablations. Subsequent task versions are specified by their actual run manifests.

Old and new PPO log probabilities use the identical saved mask. The matched ordinary Gaussian policy scores all proposals; the candidate integrates proposals that do not affect the current transition or immediate reward. Both use executed-only action cost and an entropy coefficient of zero; although the distribution implements active entropy, it does not contribute to these training losses. A fully inactive step has zero new-proposal actor score but may still train the critic. Removing a conditionally zero-mean score is a standard marginalization idea, not a new theorem: clipped PPO, advantage normalization and subsequent constraint projections preclude claims of unbiasedness or guaranteed improvement. Command clipping and IK still mean that this is not an exact density for the complete projected physical motion.

Training remains ordinary 60 Hz control-step PPO. Persistence is not, by itself, a learned Option boundary or an SMDP estimator; stage switching and re-contact initiation remain shared rules. The shared nominal controller is selected through the declared eight-case gain probe: two gains achieve 8/8 and the lower-control-step-cost setting is retained for all compared methods. This gain tuning is not assigned to the algorithm. Its status is an implemented experimental candidate whose independent value requires same-executor, matched-budget physical results and the fixed-versus-event refresh ablation. The frozen source and actual completed manifests, not this protocol description, determine which comparisons have finished.

5 Experimental Protocol

5.1 Physical tasks and evidence levels

P1 is two-stage free-cube pushing. P2 uses a 40 mm cube, a separate 30 mm research paddle and 100 mm guide rails, with a target sufficiently deep for the cube to enter the guide. It is not precision insertion of a measured industrial tray. Mass, inertia, friction, entrance direction, initial yaw and target vary through actual PhysX properties or geometry. Original robot/gripper collisions remain; the paddle's merged tool inertia retains a historical approximation.

One Isaac Lab 2.3.0 application [5] hosts independent cloned environments rather than 16 separate applications. Actions change robot references and joint drives through real physics integration; object states are not rewritten during action execution. The experiment is synthetic/state-based/actual PhysX. The ARES-R field-reference track remains separate and is neither a training-label source nor an openly published scene.

5.2 Historical splits and v7 evaluation

Historical v6 uses eight train groups, two validation groups and distinct ID, geometry and dynamics test groups. Three seeds (31, 41, 51) are evaluated on the same 32 cases per split. These historical tests have been read and cannot be reused for selecting new architecture or thresholds. The new pilot uses newly declared development groups, followed by a separately sealed final set only if the mechanism warrants promotion.

In historical v7, B0 is tuned legal damped/task-frame feedback; B1 is BODY-XY residual PPO; B2 is task-frame residual PPO; B3 is the actual effect-history candidate; B4 is finite search or scoring only when the

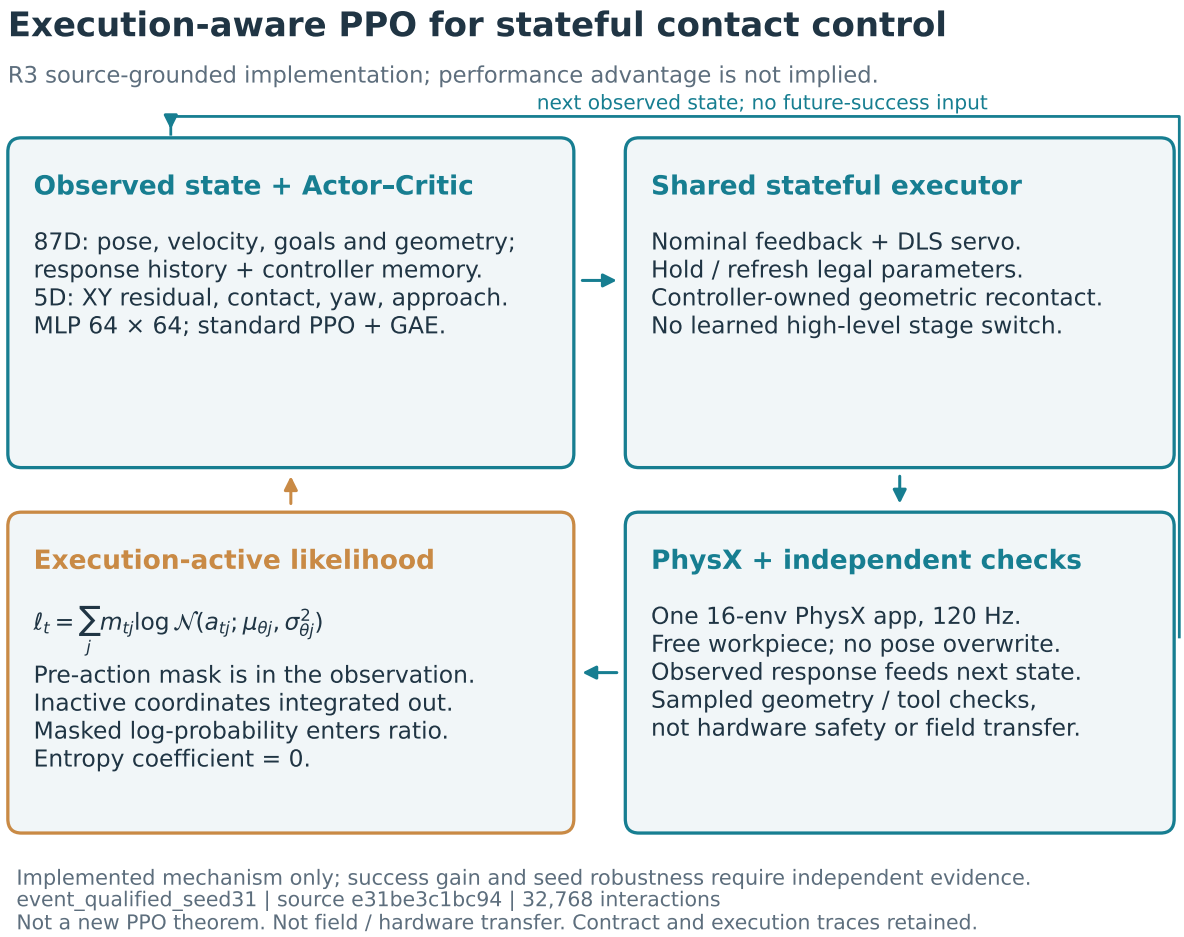


Figure 3: Actual R3 executor and actor objective, generated from the completed 87D/5D run source. The mask is known before action sampling and retained during PPO updates. Re-contact initiation and path execution are controller-owned, not a learned high-level switch. The active entropy implementation has zero weight in these runs. Implementation and a changed actor gradient are distinct from independently improved physical completion.

interface supports it. The corrected v7 pilot fixes seeds 31/41/51 and 16,384 online interactions per model, followed by 32 new P2 validation cases with case seed 102741. All learned v7 methods use the same 64-dimensional input shape, two 64-unit hidden layers and three Gaussian outputs. A failed pilot is retained, not replaced by a favorable seed. P2-only completion is the primary metric; P1, time, contact/rotation diagnostics, interaction count and planning budget are secondary.

The preregistered promotion criterion requires a positive B3–B2 P2 count difference in all three paired seeds, mean improvement of at least two cases out of 32, and conditioning interventions with a consistent behavioral/result contribution. Otherwise the mechanism remains a prototype and is not expanded to five seeds or final-test selection. A fixed $-4/0/+4$ mm phase-0 contact-parameter experiment performs three complete physical executions per case; its hindsight union is not an equal-budget online search baseline.

5.3 Independent process checks and statistics

Exact post-step terminal state is saved before reset with physics ticks. A 120 Hz sampled FCL/OBB check covers tool, cube, table and rails, plus FK/soft-limit and final stability checks. A separate mesh diagnostic samples actual robot collision geometry in a limited subset. Neither is continuous-collision detection or a hardware safety certificate. A simulated 80 N net-contact bound is uncalibrated for hardware.

We report per-seed counts and paired case differences. Seed sample standard deviation is not a binomial confidence interval, and repeated evaluation of the same cases under multiple models does not create independent environments. Time comparisons use explicitly identified common-success cases. Additional prediction/suffix interaction, demonstration continuation and diagnostic runs are accounted for separately. Software-incomplete runs are not assigned fabricated physical failure labels.

Table 1: Historical environment and evidence scope. The v7 pilot extends the actor to 64 inputs; current R1/R2/R3 interfaces are separately versioned below. Generated source: Tab1_environment.tex.

Component	Specification	Evidence boundary
Physics / control	Isaac Sim 5.1; Isaac Lab 2.3.0; 120 / 60 Hz	Verified synthetic simulation
P1 / P2	Free-workpiece two-stage push / deep guided slot	Self-built; not official benchmark
Robot / object	Single JAKA; 40 mm cube; 30 mm research paddle	No real tray grasping claim
Actor / policy	56D state-based observation; MLP 64–64	Not vision deployment
Task acceptance	Post-step state and 30 controls stable completion	Independent task/geometry checks
Formal cohort	3 methods x 3 seeds x 65,536 interactions	Historical formal training
Full historical cohort	901,120 interactions; 16 trained runs including diagnostics	Not 16 formal seeds

5.4 Subsequent closed-loop task versions and logging

The continuation uses explicitly versioned interfaces: R1 is 64D/3D with phase1 contact-target control; R2 is 66D/4D after shared rotated-contact geometry repair and finite tool yaw; formal R3 is 87D/5D after adding observable controller memory and legal re-contact. R1/R2 use a 12 s deadline, while every formal R3 comparator uses 24 s. Success thresholds are not relaxed. These task-version and horizon changes prevent attributing a cross-round count difference to the likelihood objective.

The R3 plan fixes paired seeds 31/41/51 for event-refresh ordinary and execution-active PPO, 32,768 interactions per model, plus one seed31 fixed-refresh run per objective. Completed rows and their checkpoint-linked budgets are generated from the actual run index; a scheduled but incomplete run is not evidence. R1/R2 use one seed per method and cannot support a cross-seed claim. Development cases are paired within a task version, not independent repetitions when evaluated under several seeds. Sealed final scenarios remain unopened.

Formal training saves complete 60 Hz pre-action observation/action records and episode post-step states. Its 120 Hz process traces are evaluation traces, not complete 120 Hz formal-training logs. The additional live-training showcase separately records a 44-field state sidecar at 120 Hz; it is explicitly not the complete 87D pre-action observation at that frequency. Camera re-evaluations and physical probes incur evaluation cost but no training interactions. The showcase has its own checkpoint and counted training interactions, including a failed first recording attempt; it cannot silently replace a formal model trained under the fixed comparison budget. The public package contains only synthetic task evidence, not field records.

6 Results and Research Decisions

6.1 Verified historical starting point

Table 2 reports independently accepted completions, not raw position hits. Ordinary residual PPO improves unseen-geometry aggregate completion from feedback's 25% to a seed mean of 56.25%, with sample standard deviation 15.63 percentage points. This is preliminary residual adaptation evidence, not a novel-method result. ID and dynamics results are close to feedback.

The more informative P2 breakdown is much less favorable: residual PPO obtains 6/6/7 successes out of 16, compared with feedback's 5. P1 accounts for a large share of aggregate improvement. Across common geometry successes, residual completion is 1.10–1.31 s faster depending on seed, but this is conditioned on joint success, not an unconditional task-time advantage. All models cover both tasks; a shared-training benefit has not been measured against single-task training.

6.2 Negative mechanism evidence

Paired complete physical replays compare waiting 0, 12 and 30 control steps in a legitimate handoff region before an identical suffix controller. Among 46 common valid handoffs, suffix successes are 29, 27 and 27 while average handoff speed decreases from 28.7 to 21.2 and 7.1 mm/s. Waiting alters several state variables, so velocity is not isolated causally; the experiment does not support a general waiting benefit.

Learned hold/proceed is not consistently better than ordinary residual PPO. Forcing its seed31 policy to resume preserves both the 54/64 validation count and successful-case set. A separate active-likelihood hybrid pilot changes the actual actor surrogate, but yields only 56 versus 57/64; it fails the predeclared four-case improvement threshold. These outcomes are retained as negative evidence rather than described as successful innovations.

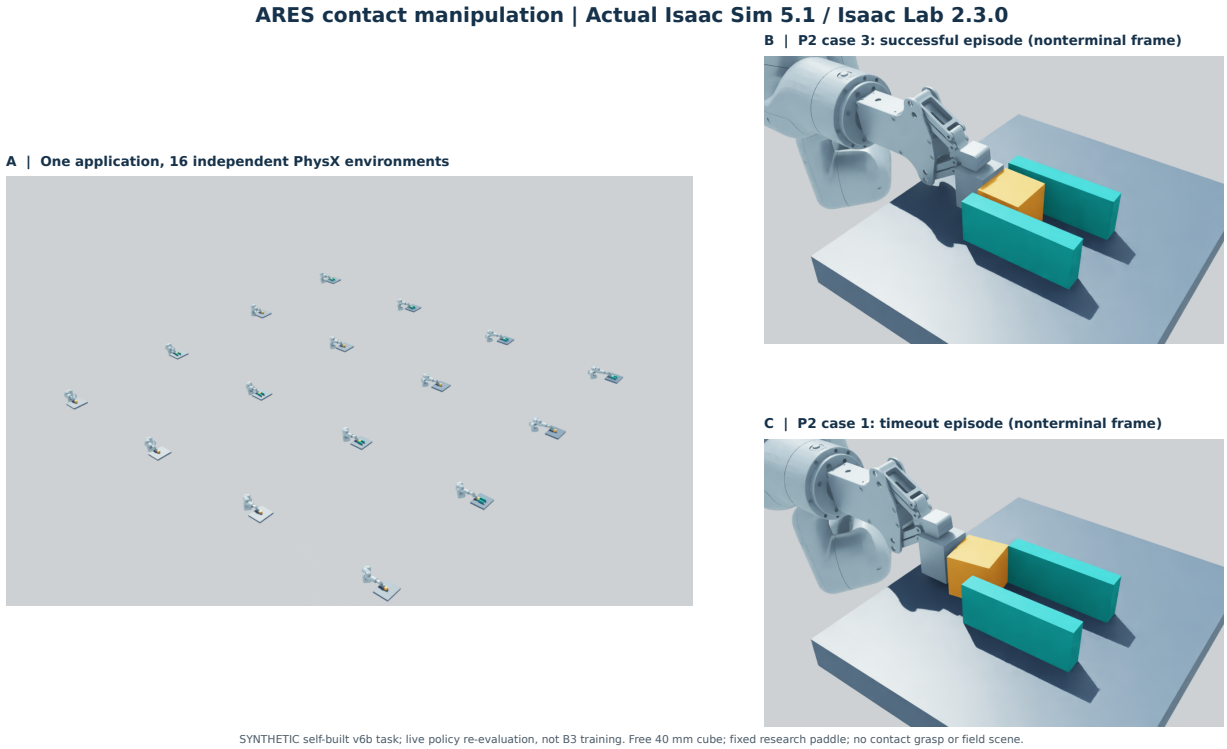


Figure 4: Actual Isaac Sim 5.1/Isaac Lab live physics evaluation of the historical v6b residual seed31 policy, not new B3 training or recorded-trajectory playback. Case3 is shown at physics tick800 (6.67 s) and later passes independent acceptance; case1 is shown at tick1184 (9.87 s) and later times out. Both images are nonterminal frames. The overview contains 16 independent environments. Physical attribute signatures are unchanged by render styling. The complete live evaluation has 55/64 independently accepted successes.

6.3 Preserved v7 development pilot

The new cohort is a development split and cannot be pooled with historical tests. Each B1/B2/B3 model receives 16,384 online interactions. Independent P2 success on 32 matched validation cases is 30/29/30 for B1, 30/30/30 for B2 and 30/30/30 for B3; corrected B0 feedback achieves 30/32. The paired B3–B2 count differences are 0/0/0. Thus the preregistered requirement of a positive difference in every seed and a mean gain of at least two cases is not met. No five-seed expansion or final-test model selection is performed. The much higher new P2 completion cannot be attributed to a novel algorithm: this cohort uses different geometry groups and a repaired shared nominal controller.

For seed31, removing all eight history channels, removing measured response while retaining command memory, or partially masking next-contract inputs preserves the same 30/32 successful-case set. Removing history changes the mean per-case applied-reference-increment RMS by about 20.6 micrometres and object-trajectory RMS by 54.2 micrometres. All compared initial robot/object/tool states match exactly in the recorded reset variables. These are real action/trajectory effects, not task-completion benefits; subsequent states are not forced to match and PhysX contact caches are not claimed to be restored. A separately trained no-next seed31 model also achieves 30/32. Fixed contact-target probes do not establish a recovery advantage and cost three full executions per case rather than one.

The new budget is 147,456 interactions for nine main models plus 16,384 for a separately trained ablation (163,840 formal), and 32,768 for two retained pre-fix software diagnostics: 196,608 new online interactions and 12 trained models in total. They are not the historical 901,120 interactions. There are 624 completed formal evaluation episodes, including training-geometry checks. Actor optimizer records show actual nonzero updates; physical trajectory and task checks are separate evidence. Persistent failures occur during phase1, approximately 29–38 mm from the goal with simulated net-contact proxies of 45–59 N, rather than only failing the final stability window.

The $-4/+4$ mm contact-target probes change mean handoff object position by approximately 1.3–1.5 mm relative to zero offset; they finish 30/29 cases versus 30 at zero. The positive offset loses one case and neither probe rescues a failure. Thus a nonzero command has some physical authority but is not a measured 4 mm displacement or a useful outcome mechanism here. A bounded existing withdrawal rule triggers three attempts on each of the two persistent failures and rescues neither; this does not prove geometric impossibility or absence of other recoveries.

Common-success mean times for B2 versus B3 are 9.3022 versus 9.1333 s, 9.4650 versus 9.2161 s, and 9.3711 versus 9.3217 s for the three seeds. The nominal controller takes 9.2461 s and the same B3 checkpoint without history takes 9.1294 s. The small B3–B2 timing difference therefore does not establish an effect-history mechanism or consistent superiority to strong feedback,

Table 2: Historical v6 independently accepted success generated from audited case records. Each split has 32 matched cases, comprising 16 P1 and 16 P2 cases; triples are seeds 31/41/51, not independent case replications. P1 cannot substitute for P2 evidence. These are not new pilot results.

Method	Task	ID	Geometry	Dynamics
Damped feedback	All	29 / 32	8 / 32	28 / 32
	P1	16 / 16	3 / 16	16 / 16
	P2	13 / 16	5 / 16	12 / 16
Direct PPO	All	0, 0, 0 / 32	0, 0, 0 / 32	0, 0, 0 / 32
	P1	0, 0, 0 / 16	0, 0, 0 / 16	0, 0, 0 / 16
	P2	0, 0, 0 / 16	0, 0, 0 / 16	0, 0, 0 / 16
Residual PPO	All	28, 28, 29 / 32	13, 18, 23 / 32	28, 29, 29 / 32
	P1	16, 16, 16 / 16	7, 12, 16 / 16	16, 16, 16 / 16
	P2	12, 12, 13 / 16	6, 6, 7 / 16	12, 13, 13 / 16
Residual + hold/proceed	All	28, 30, 30 / 32	15, 15, 18 / 32	28, 29, 29 / 32
	P1	16, 16, 16 / 16	9, 8, 12 / 16	16, 16, 16 / 16
	P2	12, 14, 14 / 16	6, 7, 6 / 16	12, 13, 13 / 16

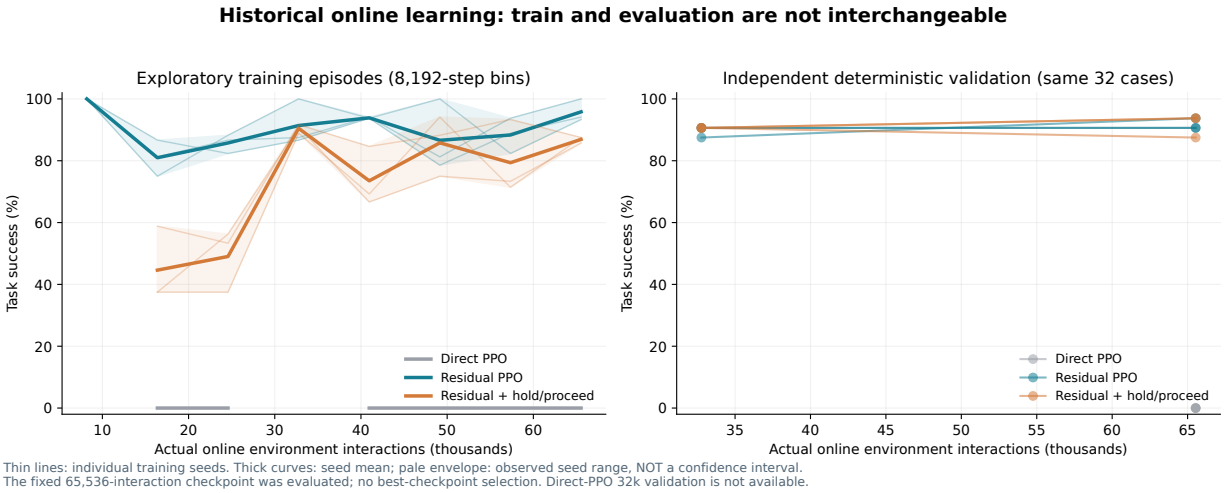


Figure 5: Actual historical online training curves. Seed traces and interaction counts are shown; training reward and frozen deterministic task success are distinct measures.

and it cannot replace the preregistered primary-success criterion.

6.4 Training correctness

The old episode-only credit prototype modified just 23.39% of seed31 samples; its boundary and all-stage modes coincided because eligible episodes ended in stage0. Corrected current-rollout bootstrapping has complete coverage but is not a stage-specific innovation. Exact post-step logging removes a terminal velocity timing ambiguity without relaxing thresholds. Final deterministic versus stochastic replay of an old model yields 51 versus 28/64, illustrating why early exploratory training episodes cannot be compared directly with a final mean policy. Direct PPO remains unreliable even after removing an unused action dimension; its zero final result is not evidence that all reasonable direct RL methods fail.

7 Subsequent Closed-Loop Research Iterations

The previous pilot is preserved as a versioned record, not replaced by this section. New runs, checkpoints, actual camera recordings and complete case tables are indexed in `08_ITERATIVE_RESEARCH`. Physical task versions and development cohorts are kept separate; results below are not new final-test scores or official benchmark replications.

7.1 R0: visual re-execution of the existing residual gain

A fresh complete physical re-execution of the historical seed41 ordinary residual policy and the feedback controller obtains 18 versus 8 accepted completions out of 32 matched geometry cases. The task decomposition is P1: 12 versus 3 out of 16, and P2: 6 versus 5 out of 16. Hence most of the aggregate gain is attributable to P1; it cannot be assigned to a new P2 method. Representative P2 case001 is a feedback failure and residual

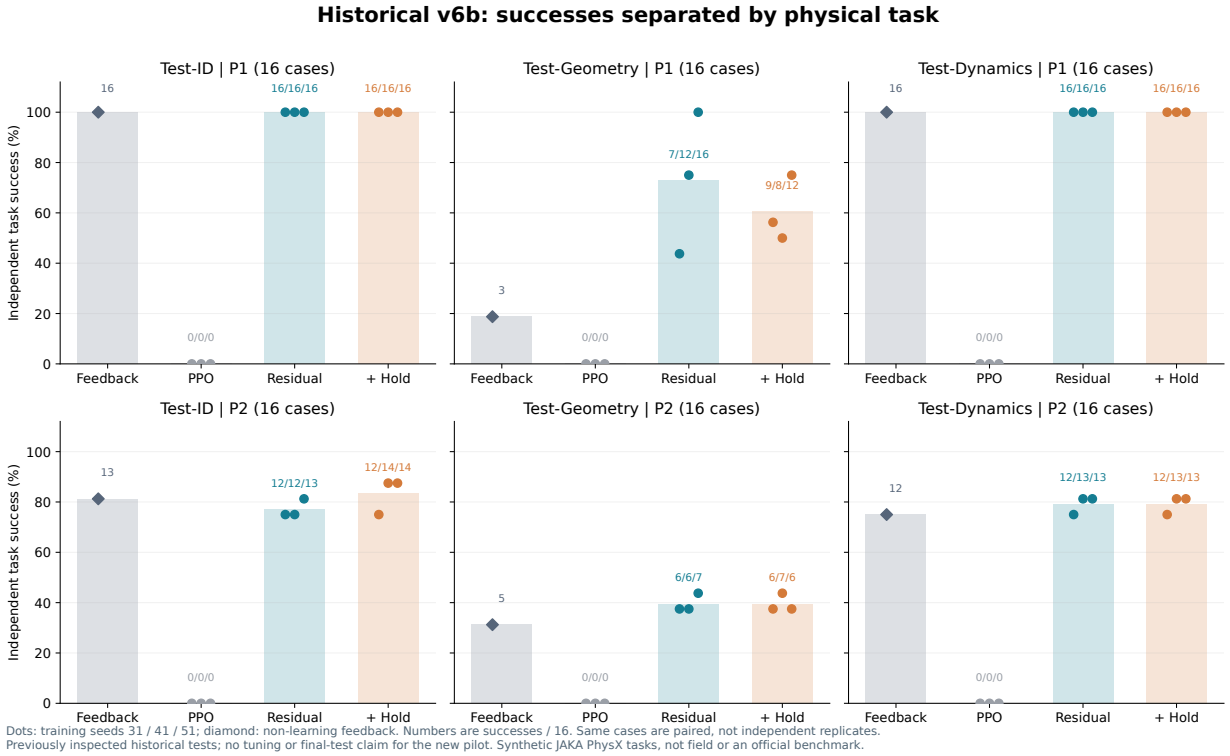


Figure 6: Historical frozen physical evaluations with per-seed and P1/P2 decomposition. The original tests are now descriptive evidence, not a new method-development validation set.

success, whereas case003 fails for both and case009 succeeds for both. These illustrative videos are selected by outcome type from the full case table, not presented as an unbiased random sample or as a novel-mechanism evaluation. Recorded robot and object initial states match; hidden PhysX solver state is not claimed to be restored.

7.2 R1: actionable contact parameter and effect history

The first new iteration gives both ordinary parameter PPO and its effect-history variant the same three-dimensional legal interface, including a phase1 contact-target offset bounded by 8 mm. Both receive 32,768 online interactions, the same 64-dimensional capacity, shared reward and physical controller; the effect-history channels are zeroed in the ordinary version. Each run completes 48 training episodes. Independent evaluation on 32 development P2 cases gives 25/32 for both, with identical successful- and failed-case sets and zero recorded tool-contract violations; same-version feedback also achieves 25/32. This development cohort differs from v7 and must not be compared by subtracting its count from v7's 30/32.

Across jointly successful cases the history policy is 0.0373 s slower, using episode control steps divided by 60 Hz rather than global simulation time. Case008 shows real action and contact differences but fails under both policies. A wider allowed parameter is not evidence that useful re-contact was learned. Observed phase1 failures motivate a shared rotated-contact footprint repair and legal tool-yaw exploration, whose engineering effects must be available to all baselines and not attributed to reinforcement-learning novelty.

7.3 R2: shared contact geometry repair and finite tool yaw

The second physical iteration shares a rotated footprint and separating-axis contact geometry repair across all controllers and gives both learning policies the same finite tool-yaw interface (66 observation and four action dimensions). The repair is an engineering correction, not a new RL mechanism. Two additional models each receive 32,768 interactions. Independent development acceptance is 24/32 for both ordinary parameter PPO and effect-history PPO, again with identical outcome sets; shared strong feedback achieves 25/32. History is 0.3083 s slower across jointly successful cases. These are separate task-version evaluations, not an ablation that attributes any R1–R2 difference to a single algorithm component.

7.4 R3: physical re-contact and execution-active likelihood

All three event-refresh seed pairs complete training and independent development audits. Ordinary PPO obtains 22/25/20 accepted successes out of 32, versus execution-active PPO's 24/24/18. Mean success is 69.79% with seed sample SD 7.86 percentage points versus 68.75% with SD 10.83 points; the paired count differences are +2/-1/-2. These are three independent training seeds on the same 32 development scenes, not 96 independent scenes. Both same-interface feedback with automatic

Table 3: Preserved v7 P2 development pilot from actual run records, not the subsequent R1–R3 task versions. Reported successes are independently accepted. Interactions denotes the underlying policy's training budget: inference interventions do not incur that training budget again. Nonlearning probes have zero training, not zero physical evaluation cost.

Method	Seed	Task	Independent success	Model train steps
B0	31	P2	30/32	0
Feedback + backoff	31	P2	30/32	0
B1	31	P2	30/32	16384
B1	41	P2	29/32	16384
B1	51	P2	30/32	16384
B2	31	P2	30/32	16384
B2	41	P2	30/32	16384
B2	51	P2	30/32	16384
B3 measured-history mask (eval)	31	P2	30/32	16384
B3 next-goal mask (eval)	31	P2	30/32	16384
B3 no next goal (train)	31	P2	30/32	16384
B3 history mask (eval)	31	P2	30/32	16384
B3	31	P2	30/32	16384
B3	41	P2	30/32	16384
B3	51	P2	30/32	16384
Fixed contact +4 mm	31	P2	29/32	0
Fixed contact -4 mm	31	P2	30/32	0

Table 4: Mechanism interventions and resource accounting generated from the actual pilot evidence. An input or action effect is not by itself evidence of improved completion.

Variant	Frame	Result/command history	Next goal	Seeds	Train budget
B0	Task	Zero / none	Observed	31	0
Feedback + backoff	Task	Zero / none	Observed	31	0
B1	Body	Zero / none	Observed	31,41,51	16384
B2	Task	Zero / none	Observed	31,41,51	16384
B3	Task	Observed	Observed	31,41,51	16384
B3 no measured effect (eval)	Task	Command memory only	Observed	31	Same checkpoint
B3 no next goal (train)	Task	Observed	Hidden during training	31	16384
B3 no next goal (eval)	Task	Observed	Suppressed at eval	31	Same checkpoint
B3 no history (eval)	Task	Suppressed at eval	Observed	31	Same checkpoint
Fixed contact +4 mm	Task	Zero / none	Observed	31	0
Fixed contact -4 mm	Task	Zero / none	Observed	31	0

re-contact and the best nominal feedback without it achieve 26/32. Each active-policy successful-case set is a subset of feedback's, with no rescue and 2/2/8 regressions. Neither improved completion nor reduced seed variability is supported.

For seed31, active PPO rescues three ordinary-PPO failures but regresses on one, a genuine +2-case local signal. It does not replicate across seeds. The all-case mean duration falls 0.7474 s for that seed, but the 21 jointly successful cases are 0.3325 s slower. The corresponding common-success differences are also slower by 0.6632 and 0.3139 s for seeds41/51. Against strong feedback, the active policy is slower on shared successes by 0.227/0.560/0.060 s. Thus selected rescues or a favorable all-case mean cannot establish a general time advantage.

The fixed-refresh seed31 ablation gives 25/32 for both ordinary and active likelihood, with identical accepted-case sets. Compared with event refresh's 22/24, this does not support a benefit of the contact-event heuristic. On the 25 common fixed-refresh successes, the active version is 0.208 s faster. This is retained as a real single-seed secondary-cost signal, not a repeatable primary-success advantage or a cross-seed method conclusion. The four cells share the executor, 24 s contract and training budget; the event/fixed comparison is not mixed with R1/R2's shorter task version.

The new formal training comprises twelve models at 32,768 interactions each, or 393,216 interactions, including both fixed-refresh ablations. Two software/optimizer smokes add 5,120, a separate completed live-training capture adds 12,288, and its retained failed recording attempt adds 4,096 interactions: 414,720 charged online interactions in total. There are 15 completed training checkpoints, not 15 formal comparison seeds. The failed attempt reaches its first rollout then fails in camera-wrapper serialization before an optimizer update; its known work is charged, while unknown reset-related physics cost is not invented. Physical probes and camera evaluations are separately indexed, not new independent training scenes.

Exploratory no-guide pushing transfer. Removing the guide rails provides a physically different contact condition without additional policy training. On 32 exploratory cases, feedback and the ordinary seed31 pol-

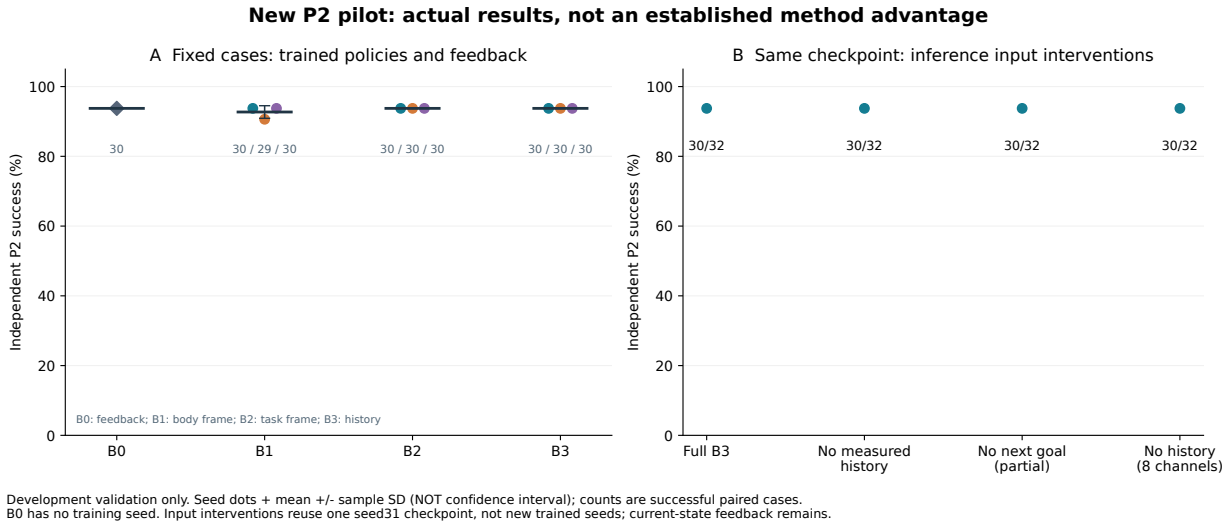


Figure 7: Preserved v7 P2 development evaluation and same-checkpoint conditioning interventions. B3 does not improve over B2. Mask interventions are evaluations of one trained model, not additional training seeds. This is not an R1–R3 result.

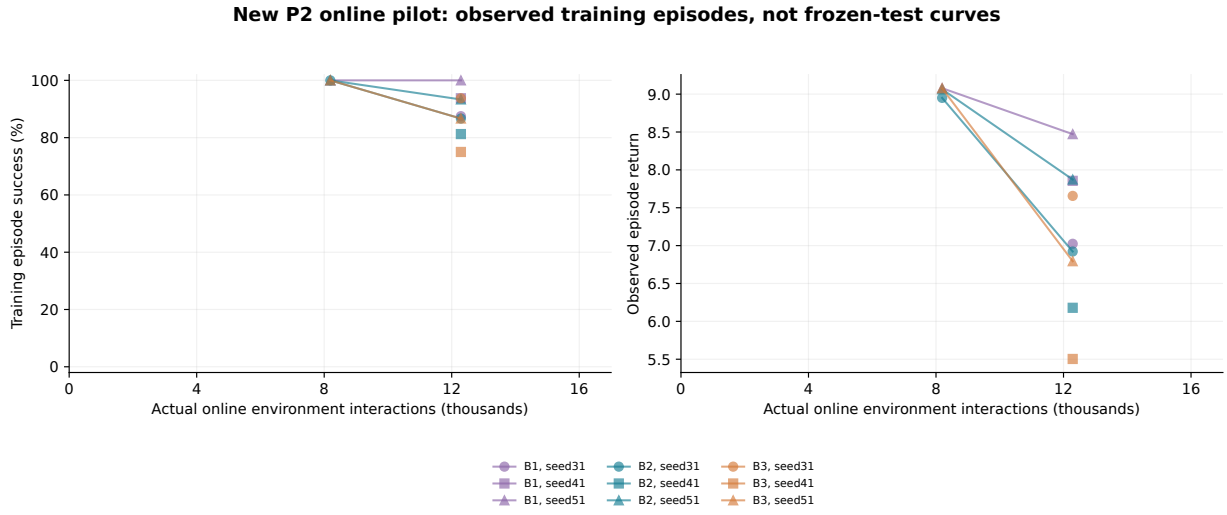


Figure 8: Preserved v7 online episode curves under its fixed pilot budget. Exploratory episode reward/success is distinct from independent validation. Later R1–R3 interaction budgets are not included here.

icy each pass 32/32, while the execution-active seed31 policy passes 29/32 and is 2.588 s slower on common successes. This is a zero-training transfer probe, not a sealed final test or evidence of shared-policy improvement. It does not turn the historical P1/P2 results into a verified cross-task benefit of the new mechanism.

Real-rollout actor-gradient diagnostic. The formal seed31 ordinary and execution-active runs save exactly equal first 512 observations, sampled actions, GAE values and execution masks, with exactly equal pre-update policy state dictionaries. On these real on-policy samples, a same-weight diagnostic changes the score-gradient vector by Euclidean norm 0.405706 with cosine 0.933645; old log-probability reconstruction error is below 1.2×10^{-7}. It consumes no extra environment interactions or optimizer steps. This isolates the score construction on the inspected samples, not the entire first rollout or a performance gain. The diagnostic uses its own common advantage normalization, and is not substituted for actual training minibatch updates.

For that seed's full 32,768 training interactions, the ordinary policy enters automatic re-contact for 576 control samples and has only three valid normal-angle samples; the execution-active policy has neither. Thus the shared executor is implemented, but it is false to claim the active policy learned recovery or meaningful use of every new action dimension. The completed multi-seed physical results, not a nonzero actor gradient, will determine the mechanism decision.

8 Limitations and Conclusion

The study establishes actual parallel contact-RL training, a limited historical residual-control geometry result, and a correctly connected execution-aware actor objec-

Table 5: Completed new-round P2 development evaluations generated from the actual run index. Independent success is distinct from raw reward. Seeds refer to independently trained policies, not independent replications of the same test scenes. Each round has a separately versioned physical interface and duration contract, so cross-round differences are not a single-component causal ablation. Camera re-executions, training-geometry checks and incomplete audits are excluded.

Round	Method	Seed	Obs./act.	Train interactions	Independent success
R1	Feedback	–	64/3	0	25/32
R1	Parameter PPO	31	64/3	32768	25/32
R1	Effect-history PPO	31	64/3	32768	25/32
R2	Feedback	–	66/4	0	25/32
R2	Yaw parameter PPO	31	66/4	32768	24/32
R2	Yaw + history PPO	31	66/4	32768	24/32
R3	Feedback / nominal	–	66/4	0	26/32
R3	Re-contact feedback	–	87/5	0	26/32
R3	Event / ordinary PPO	31	87/5	32768	22/32
R3	Event / ordinary PPO	41	87/5	32768	25/32
R3	Event / ordinary PPO	51	87/5	32768	20/32
R3	Event / active PPO	31	87/5	32768	24/32
R3	Event / active PPO	41	87/5	32768	24/32
R3	Event / active PPO	51	87/5	32768	18/32
R3	Fixed / ordinary PPO	31	87/5	32768	25/32
R3	Fixed / active PPO	31	87/5	32768	25/32

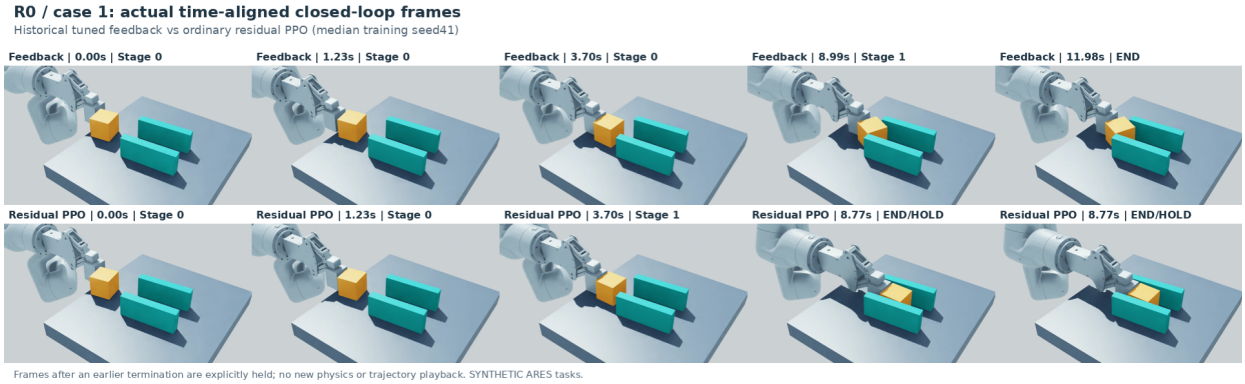


Figure 9: Fresh R0 closed-loop PhysX evaluation of historical P2 case001: feedback fails and ordinary residual PPO passes independent qualification. The same physical clocks are sampled in both rows; END/HOLD explicitly denotes a completed episode held for comparison. This is an existing-policy example, not the new candidate method.

tive. It does not establish a superior new manipulation algorithm. R1/R2 effect-history policies do not improve accepted completions; R3's paired three-seed differences are +2/-1/-2 cases and its sample standard deviation is larger, not smaller. The first favorable seed cannot support a robustness claim. All three execution-active successful-case sets are subsets of shared feedback successes: there are no rescues against that controller, only 2/2/8 regressions. Gains in one selected video therefore cannot replace the full comparison.

The score-gradient experiment checks only 512 saved real samples and corresponding initial weights, not a policy-gradient theorem or all contact solver state. The formal seed31 execution-active policy never triggers re-contact and has no valid normal-angle samples, while the ordinary policy has only three. Thus the extended recovery interface is not evidence that recovery or every proposed parameter was learned. Better use of rare legal parameter opportunities, sufficient training coverage and a physically useful parameter intervention must be established before another scaling run. Negative results under the current budget do not disprove contact RL, execution history or marginalization in general.

The v7 near-ceiling pilot, the R1/R2 12 s interfaces and the R3 24 s executor are distinct task versions. Shared footprint repairs, nominal-gain tuning and additional time cannot be attributed to a new RL objective. Repeated development evaluation selects research mechanisms; no sealed-final-set generalization claim is justified. Three seeds and a short pilot budget are insufficient to establish convergence or robust differences. Common-success time excludes failures and is reported separately from all-case cost. Repeated evaluation of the same 32 scenes is not 96 independent scenes.

State-based simulator observations include exact pose and known dynamics context. A deployed perception system, visual/point-cloud actor, measured tool dynamics, true grasping, full robot CCD and real-robot validation have not been demonstrated. The synthetic cube and research paddle are not the field tray. Independent 120 Hz evaluation audits and the separate showcase state sidecar are not complete 120 Hz formal-training observations or a hardware-safety certificate. The re-contact helper avoids the planar object footprint but is not a full

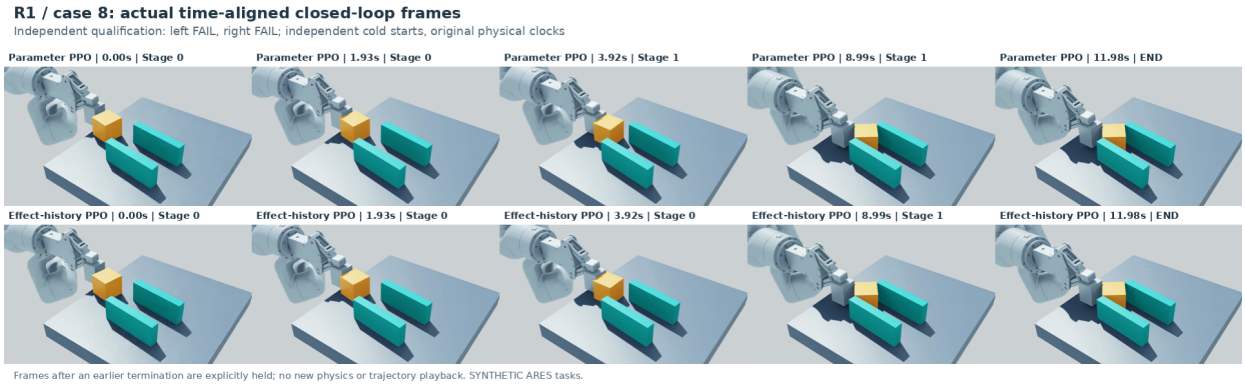


Figure 10: R1 case008, actual same-case executions after new online training. Both strategies fail despite different object/contact trajectories. The complete 32-case comparison, rather than this example alone, supports the equal 25/32 outcome.

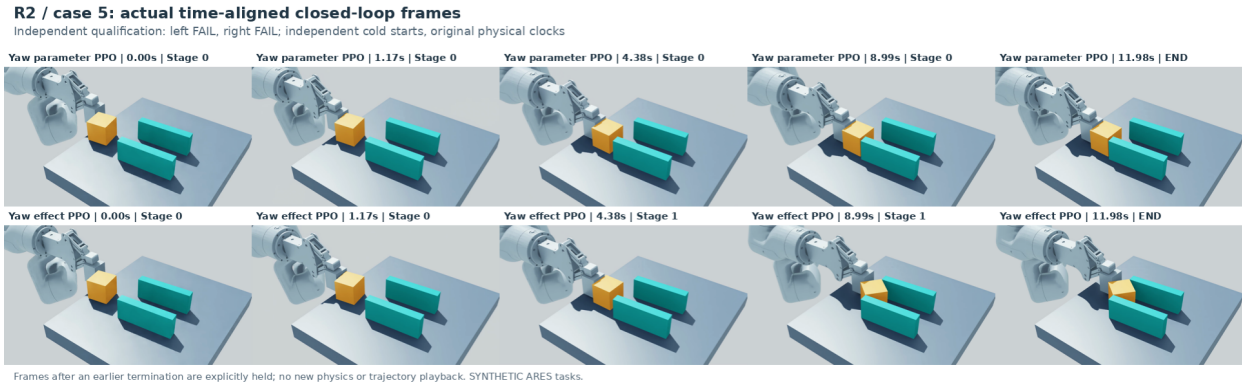


Figure 11: Actual R2 P2 case005 closed-loop comparison after rotated geometry and finite tool-yaw repair. Both policies fail. Their initial recorded physical states match, while contact trajectories differ. The shared engineering changes do not demonstrate an RL method contribution.

robot/rail motion planner. Field references remain private and separate. Long-horizon learned options, VLA instructions and general skill recovery remain extensions, not completed contributions.

The next bounded research decision is to retain the strong controller and ordinary residual policies, test usable contact/re-contact authority on actual failures, and select a parameter interface that changes useful physical outcomes before further method expansion. Event refresh and execution-active likelihood are not promoted on the present evidence. Only a repeatable benefit with mechanism ablations would warrant five seeds, stronger related-method controls and sealed geometry/dynamics tests. The current package is a reproducible research manuscript and client evidence deliverable, not a claim of SCI submission readiness. This narrows the method hypothesis without discarding the platform or the requirement to investigate reinforcement learning.

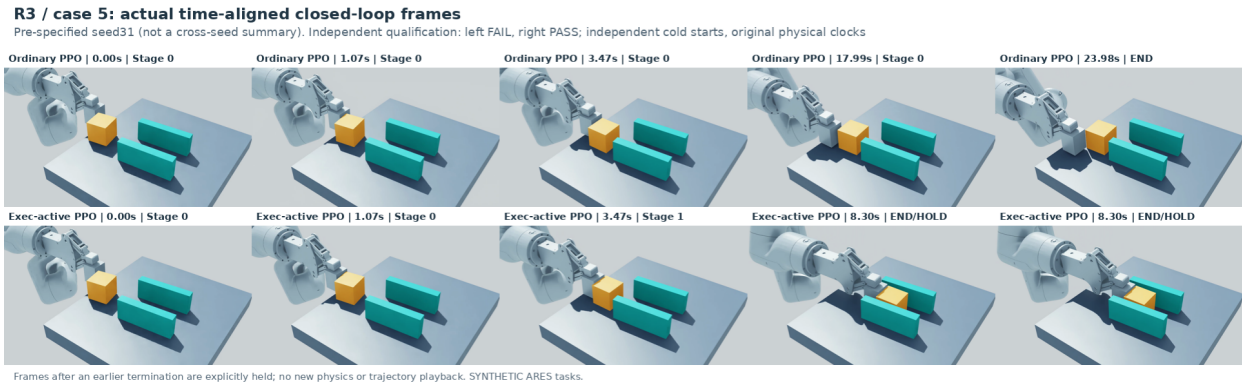


Figure 12: Actual seed31 P2 case005, an ordinary-policy failure and execution-active success. This represents one of three rescues and one regression in that seed, not a repeatable three-seed advantage. No learned recovery is inferred: the active training run has no re-contact activation. All case outcomes and the stronger feedback comparison remain reported.



Figure 13: Actual additional online-training capture: sixteen camera views of independent environments in one Isaac application at the same physics tick1536, not a single-camera panorama or sixteen applications. The run performs 12,288 additional interactions, three rollout updates and 288 optimizer minibatches. It is not one of the fixed-budget formal performance models. Full video, a separate true wide-camera view and before/after optimizer-state evidence are included.

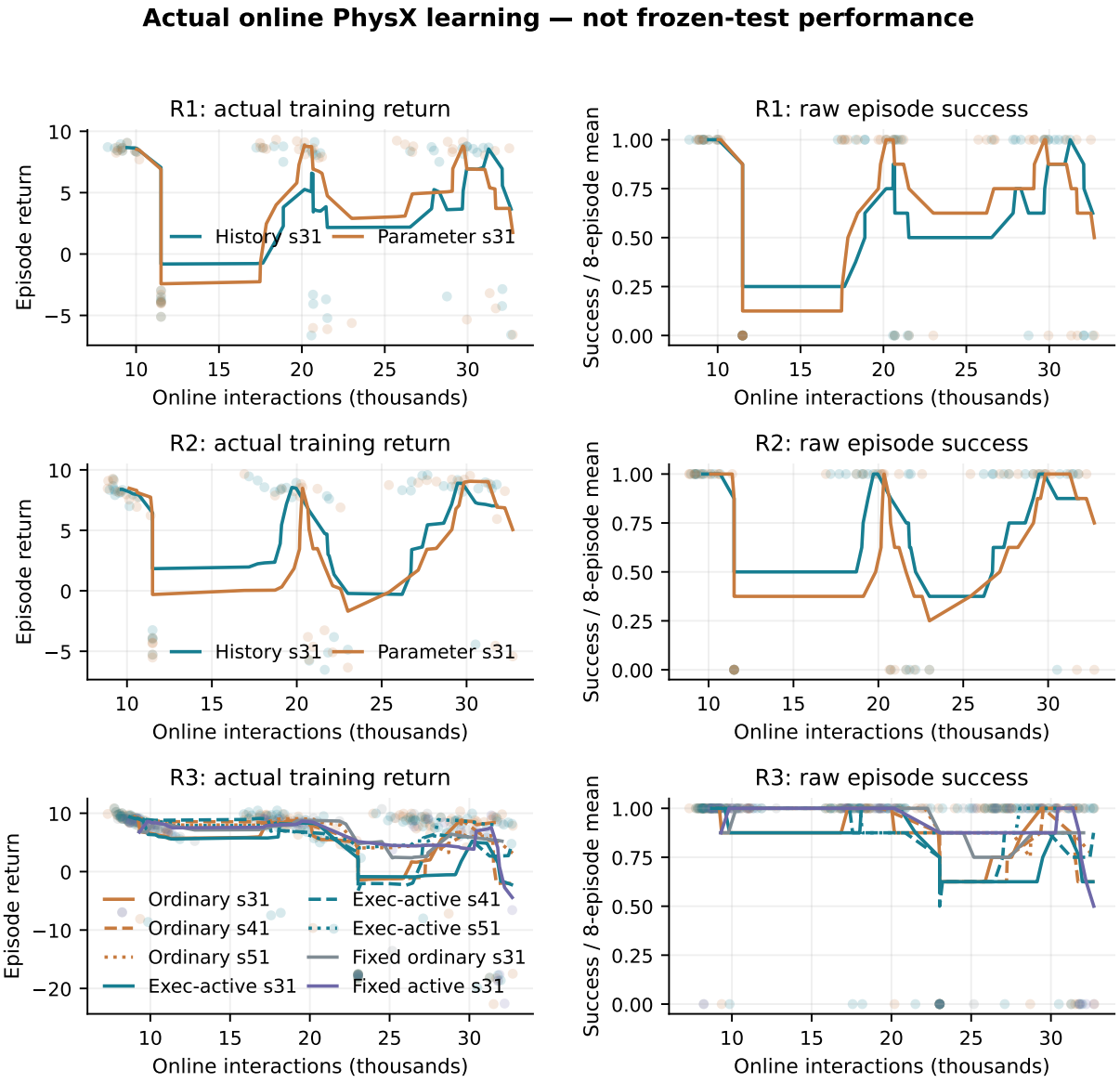


Figure 14: Actual completed R1–R3 online episode records. Rewards and training episode outcomes are not independently accepted development performance. Model seeds and physical task versions are distinguished; software probes and showcase continuation do not inflate the formal curves.

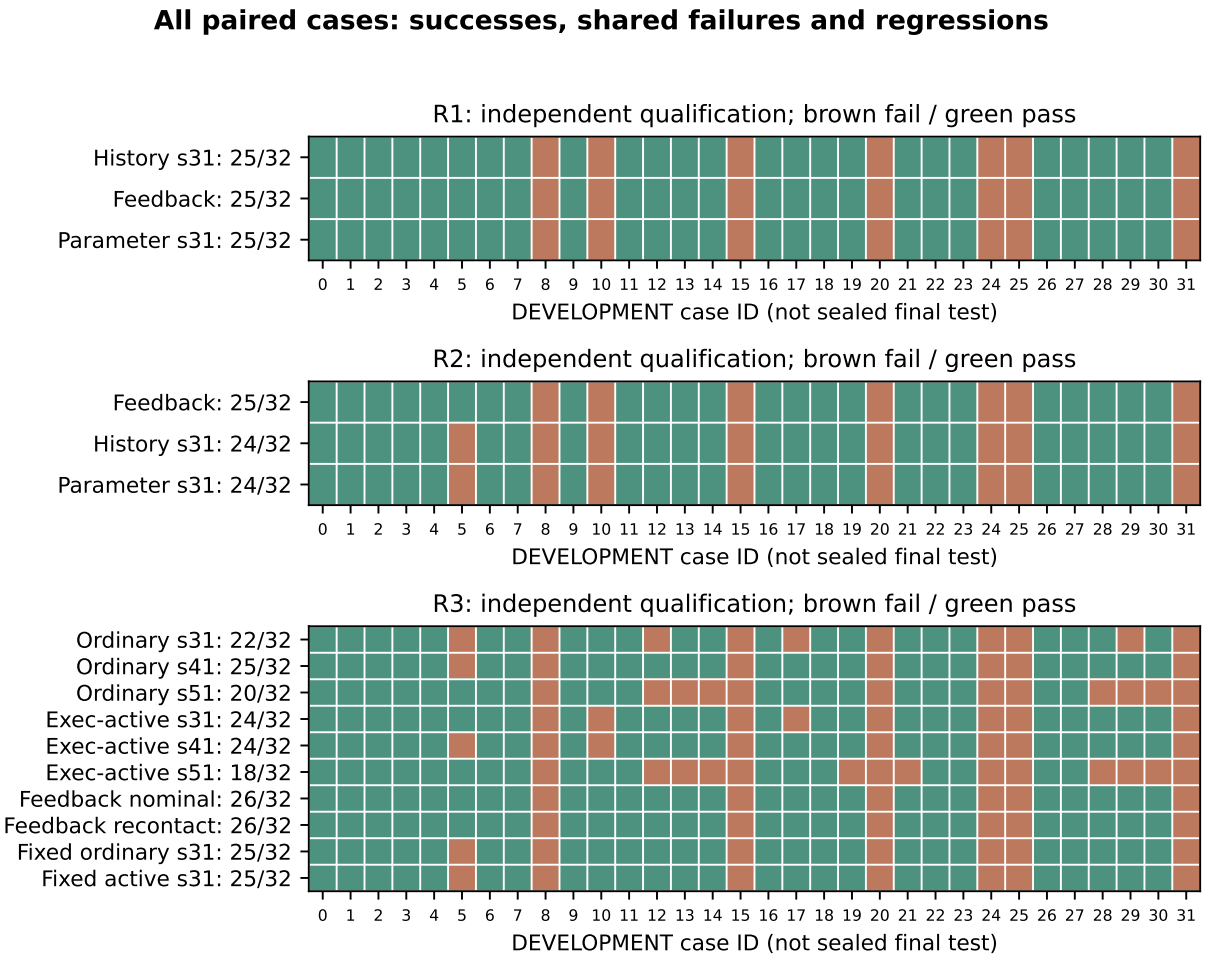


Figure 15: All matched development-case outcomes, including shared failures, rescues and regressions. R1/R2 and R3 use separately versioned interfaces and horizons. The complete case matrix, not a selected filmstrip, supports the method decision.

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